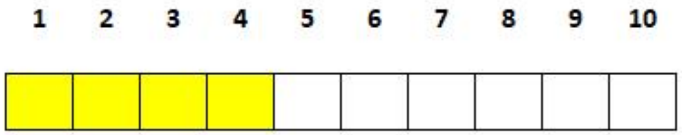
```
Empty = 6
Full  = 4
Mutex = 1
```

1	2	3	4	5	6	7	8	9	10

```
#define N 100                    /* number of slots in the buffer */
typedef int semaphore;           /* semaphores are a special kind of int */
semaphore mutex = 1;             /* controls access to critical region */
semaphore empty = N;             /* counts empty buffer slots */
semaphore full = 0;              /* counts full buffer slots */

void producer(void)
{
    int item;

    while (TRUE) {               /* TRUE is the constant 1 */
        item = produce_item();   /* generate something to put in buffer */
        down(&empty);            /* decrement empty count */
        down(&mutex);            /* enter critical region */
        insert_item(item);       /* put new item in buffer */
        up(&mutex);              /* leave critical region */
        up(&full);               /* increment count of full slots */
    }
}
```

```
void consumer(void)
{
    int item;

    while (TRUE) {               /* infinite loop */
        down(&full);             /* decrement full count */
        down(&mutex);            /* enter critical region */
        item = remove_item();    /* take item from buffer */
        up(&mutex);              /* leave critical region */
        up(&empty);              /* increment count of empty slots */
        consume_item(item);      /* do something with the item */
    }
}
```

```
Empty = 6
Full  = 4
Mutex = 1
```



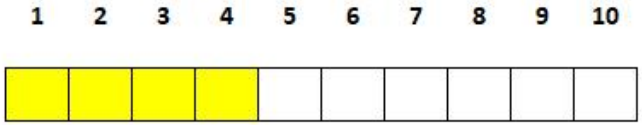

```
Empty = 6
Full  = 4
```

```
Empty = 6
Full  = 4
```

```
void producer(void)
{
        int item;

        while (TRUE) {
                item = produce_item( );
                down(&empty);
                down(&mutex);
                insert_item(item);
                up(&mutex);
                up(&full);
        }
}
```

```
void consumer(void)
{
        int item;

        while (TRUE) {
                down(&full);
                down(&mutex);
                item = remove_item( );
                up(&mutex);
                up(&empty);
                consume_item(item);
        }
}
```

Empty = 6
Full = 4
Mutex = 1



Empty = 5
Full = 4

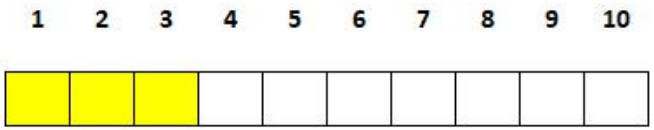
Empty = 5
Full = 4

```
void producer(void)
{
        int item;

        while (TRUE) {
                item = produce_item( );
                down(&empty);
                down(&mutex);
                insert_item(item);
                up(&mutex);
                up(&full);
        }
}
```

```
void consumer(void)
{
        int item;

        while (TRUE) {
                down(&full);
                down(&mutex);
                item = remove_item( );
                up(&mutex);
                up(&empty);
                consume_item(item);
        }
}
```

Mutex = 1

```
Empty = 6
Full  = 4
Mutex = 1
```




```
Empty = 5
Full  = 3
```

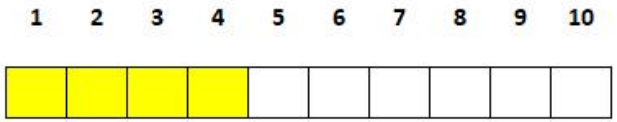
```
Empty = 5
Full  = 3
```

```
void producer(void)
{
        int item;

        while (TRUE) {
                item = produce_item( );
                down(&empty);
                down(&mutex);
                insert_item(item);
                up(&mutex);
                up(&full);
        }
}
```

```
void consumer(void)
{
        int item;

        while (TRUE) {
                down(&full);
                down(&mutex);
                item = remove_item( );
                up(&mutex);
                up(&empty);
                consume_item(item);
        }
}
```

```
Mutex = 0

Mutex = 1
```

```
Empty = 6
Full  = 4
Mutex = 1
```




```
Empty = 5
Full  = 3
```

```
Empty = 5
Full  = 3
```

```
void producer(void)
{
        int item;

        while (TRUE) {
                item = produce_item( );
                down(&empty);
                down(&mutex);
                insert_item(item);
                up(&mutex);
                up(&full);
        }
}
```

Mutex = 0

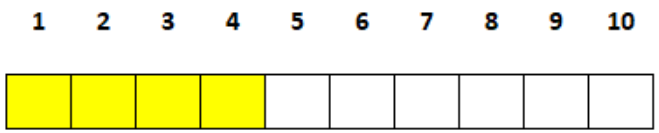
Mutex = 1

```
void consumer(void)
{
        int item;

        while (TRUE) {
                down(&full);
                down(&mutex);
                item = remove_item( );
                up(&mutex);
                up(&empty);
                consume_item(item);
        }
}
```

Mutex = 0

Mutex = 1

```
Empty = 6
Full  = 4
Mutex = 1
```




```
Empty = 6
Full  = 3
```

```
Empty = 6
Full  = 3
```

```
void producer(void)
{
        int item;

        while (TRUE) {
                item = produce_item( );
                down(&empty);
                down(&mutex);
                insert_item(item);
                up(&mutex);
                up(&full);
        }
}
```

Mutex = 0

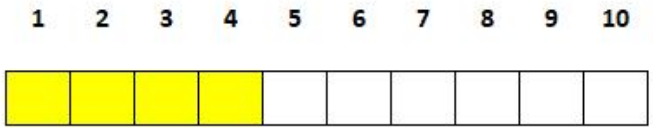
Mutex = 1

```
void consumer(void)
{
        int item;

        while (TRUE) {
                down(&full);
                down(&mutex);
                item = remove_item( );
                up(&mutex);
                up(&empty);
                consume_item(item);
        }
}
```

Mutex = 0

Mutex = 1

Empty = 6
Full = 4
Mutex = 1



Empty = 6
Full = 4

Empty = 6
Full = 4

```
void producer(void)
{
    int item;

    while (TRUE) {
        item = produce_item( );
        down(&empty);
        down(&mutex);
        insert_item(item);
        up(&mutex);
        up(&full);
    }
}
```

Mutex = 0

Mutex = 1

```
void consumer(void)
{
    int item;

    while (TRUE) {
        down(&full);
        down(&mutex);
        item = remove_item( );
        up(&mutex);
        up(&empty);
        consume_item(item);
    }
}
```

Mutex = 0

Mutex = 1